Examiner only

10. A technician prepared 250 cm^3 of a 0.50 mol/dm^3 solution of citric acid, $C_6H_8O_7$.

(a) Calculate the number of moles of citric acid in the solution. [2]

$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{number of moles}}{\text{volume (dm}^3\text{)}}$$

$$1\,dm^3 = 1000\,cm^3$$

Number of moles = ... mol

(b) Calculate the mass of citric acid in the solution. [2]

$A_r(H) = 1$ $A_r(C) = 12$ $A_r(O) = 16$

$$\text{number of moles} = \frac{\text{mass (g)}}{M_r}$$

Mass = ... g

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END OF PAPER